PHASE H

v2.1 implementation plan

Edge consolidation + filter changes + Kelly sizing
+ multi-fire + compounding
POLYMARKET 5-minute BTC up/down bot · 2026-05-13

What this plan does: pulls every confirmed edge from the bot's 5-day production history + Bonereaper studies + Phase G analysis, eliminates the 6 confirmed losers, exploits the 5 confirmed winners, and adds the 3 architectural improvements (Kelly + multi-fire + compounding) into a single coherent v2.1 strategy.

Projection (Phase G Conservative scenario): $1,006 bankroll grows to $116k median by day 90 if the v2.1 cell-level edge (58.2% WR backtest) generalizes.

1. Executive summary

Phase G uncovered that the bot has been firing in losing cells. v2's filters eliminate the big losers but miss the biggest winner: **tier 30-50% × |Δ_bps|<5** (n=83, +$0.65/trade, 48% WR with asymmetric payoff). v2 currently rejects this zone via its global |Δ|≥5 floor.

v2.1 is v2 + 4 filter changes (CH1-CH4) + Kelly sizing (CH5-CH6) + multi-fire (CH7-CH8). All changes have data backing from Phase G or earlier studies. Implementation is ~150 LOC + 15 tests.

Recommended Phase H deployment sequence:
H.1 — this PDF (now)
H.2 — code (next session, ~1h)
H.3 — tests (next session, ~30min)
H.4 — register + .envrc edit (next session, ~5min)
H.5 — VPS deploy + 3-stage verify (next session, ~1h)
H.6 — 7-day soak gate (+7d after H.5) → PROMOTE/EXTEND/ROLLBACK

2. Edge inventory (full bot history)

2.1 Confirmed winners (statistically supported)

#	Edge	Source	n	WR / per-trade	Status
W1	Tier 30-50% × |Δ|<5	Phase G §5.1	83	48.1% / +$0.65	★ NEW — exploit
W2	Tier 70-85% (any |Δ|)	Phase G §4.1 + edge-disc 5/9	95	70.6%	v2 allows ✓
W3	Regime = trending	regime-validation 5/10	19	70.59% / +$1.75	v3 has gate; v2.1 may add
W4	EV ∈ [0.05, 0.20)	Phase G §4.4	54	72% / +$1.61	v2 allows ✓ (but allows losers too)
W5	Δ ≠ 0 (any sign)	Phase G §4.3	446	51.6%	v2 allows ✓

2.2 Confirmed losers (Wilson CI confirms)

#	Edge	Source	n	WR / per-trade	Status
L1	Δ_bps = 0 (no skew)	Phase G §4.3	187	44.6% / –$0.40	✗ v2.1: REJECT
L2	EV ∈ [0.02, 0.05)	Phase G §4.4	77	36.0% / –$1.46	✗ v2.1: REJECT bucket
L3	Tier 50-70%	edge-disc 5/9 + Phase G	349	50.4% / –$0.25	✓ v2 already rejects
L4	Tier 85-100% (slow-min)	scout-final 5/8	n/a	–$3,964/7d Boneread	✓ v2 already rejects
L5	wide_fire + ensemble	Phase F audit	126	42.2% / –$0.56	✓ Phase F cut removed
L6	AGGRESSIVE_MODE	May 8/9 split	299	high variance	✓ already off

2.3 Suggestive (n<30 or marginal — DEFER)

#	Edge	Source	Note
S1	Hour 0/17/21 > 5/14/23 UTC	Phase G §4.6	underpowered, defer
S2	spot_dir UP_weak > DOWN_weak	Phase G §5	n thin, defer

2.4 Architectural (Phase G additions)

#	Edge	Source	Status in v2.1
N1	Kelly sizing	Phase G §10.1-10.2	✓ implement (CH5)
N2	Multi-fire per window	Phase G §10.3	✓ implement (CH7-CH8)
N3	Cumulative compounding	Phase G §10.4	✓ implicit via Kelly + bankroll
N4	Tier-specific Δ floor	Phase G §5.1	✓ implement (CH3)

3. Decision matrix — what's in v2.1, what's deferred

Edge	v2 currently	v2.1 change	Expected impact
W1 — 30-50%×|Δ|<5	REJECTED (|Δ|≥5)	ALLOW (CH3)	+ ~16 fires/day in cell
W2 — 70-85% any Δ	ALLOWED	unchanged	—
W3 — trending regime	no gate	OPTIONAL flag	→ operator Q2
W4 — EV [0.05,0.20)	ALLOWED	unchanged	—
W5 — Δ ≠ 0	ALLOWED	REJECT Δ=0 (CH1)	– 187 losing fires removed
L1 — Δ = 0	ALLOWED	REJECT (CH1)	+$75 PnL recoverable
L2 — EV [0.02,0.05)	ALLOWED	REJECT (CH2)	+$112 PnL recoverable
L3 — Tier 50-70%	REJECTED	unchanged ✓	—
L4 — Tier 85-100%	REJECTED	unchanged ✓	—
L5 — wide_fire	REMOVED	unchanged ✓	—
L6 — aggressive	OFF	unchanged ✓	—
S1 — hour gate	no gate	DEFER	needs more data
S2 — spot_dir gate	no gate	DEFER	needs more data
N1 — Kelly sizing	$5 flat	Kelly (CH5)	compounding growth
N2 — multi-fire	1/window	2/window (CH7-8)	2× fires/day
N3 — compounding	none	implicit via Kelly	geometric growth
N4 — tier Δ floor	global 5 bps	tier-specific (CH3)	exploits W1

4. v2.1 architecture

v2.1 inherits from SpotAwareStrategyV2 and overrides _maybe_fire with the new filter chain, plus adds a get_size method for Kelly and lifts the once-per-window invariant.

4.1 Filter chain (per-fire decision)

```
class SpotAwareStrategyV2_1(SpotAwareStrategyV2):
    def _maybe_fire(self, *, tier, p_win, ev, delta_bps, side, ...):
        # CH1 — reject Δ=0
        if delta_bps == 0.0: return None
        # CH2 — reject EV in [0.02, 0.05)
        if 0.02 <= ev < 0.05: return None
        # CH3 — tier-specific Δ floor
        min_d = 0.0 if tier in ('15-30%', '30-50%') else 5.0
        if abs(delta_bps) < min_d: return None
        # v2's tier rejection (50-70% + 85-100%) inherited
        return super()._maybe_fire(...)
```

4.2 Kelly sizing (per-fire bet)

```
@dataclass(frozen=True, slots=True)
class KellyConfig:
    kelly_mult: float = 0.10 # fractional Kelly (start conservative)
    min_bet_abs: float = 1.0
    max_bet_abs: float = 500.0 # Polymarket retail liquidity
    max_bet_pct: float = 0.10 # never more than 10% of bankroll

def kelly_bet(p_win, price, bankroll, cfg):
    f = max(0, (p_win - price) / (1 - price))
    bet = bankroll * cfg.kelly_mult * f
    bet = min(bet, bankroll * cfg.max_bet_pct, cfg.max_bet_abs)
    return max(cfg.min_bet_abs, bet) if f > 0 else 0
```

4.3 Multi-fire per window

v2.1 tracks fires_this_window instead of fired_for_window. Allows up to MULTI_FIRE_CAP (default 2) fires per 5-min window. Caller (PaperRunner) resets at window boundary.

Risk note: multi-fire increases capital deployed per window. With Kelly 0.10× × max 2 fires × ~5% avg bet = ~10% of bankroll per window peak exposure. Sane.

4.4 Compounding (live bankroll)

v2.1 constructor takes a bankroll_getter callback. Each fire reads current bankroll from PaperRunner, sizes accordingly. As bankroll grows, bet sizes grow proportionally (clamped by max_bet_abs). This is the geometric compounding that drives the Phase G $116k/90d projection.

5. Implementation sub-phases

Sub-phase	When	Deliverable	Effort
H.1	this session	PDF (this doc)	~30 min
H.2	next session	src/.../spot_aware_v2_1.py	~1 h
H.3	next session	tests/.../test_spot_aware_v2_1.py	~30 min
H.4	next session	STRATEGY=spot_aware_v2_1 registered + .envrc	~5 min
H.5	next session	VPS deploy + +5/30/24h verify	~1 h
H.6	+7d after H.5	soak gate → PROMOTE / EXTEND / ROLLBACK	passive

5.1 H.2 code skeleton

File: src/polymarket_bot/strategies/spot_aware_v2_1.py (~150 LOC)
Type hints + module docstring (≥80 chars) per anti-vibe rule #5/#6.
Inherits SpotAwareStrategyV2.
Adds: KellyConfig, bankroll_getter callback, fires_this_window counter.
Overrides: _maybe_fire (filter chain), get_size (Kelly).
Lifts: fired_for_window invariant.

5.2 H.3 test surface (~15 tests)

Test	What it validates
test_v2_1_rejects_delta_zero	CH1 — Δ=0 returns None
test_v2_1_rejects_ev_in_bad_bucket	CH2 — EV ∈ [0.02, 0.05) returns None
test_v2_1_allows_tier_30_50_with_low_delta	CH3 — tier 30-50% × |Δ|=2 fires
test_v2_1_requires_5bps_for_tier_70_85	CH3 — tier 70-85% × |Δ|=2 returns None
test_v2_1_inherits_tier_50_70_reject	v2 inheritance — 50-70% returns None
test_kelly_bet_basic	CH5 — bet math for known inputs
test_kelly_bet_clamps_to_max_abs	CH5 — never exceeds $500
test_kelly_bet_clamps_to_max_pct	CH5 — never exceeds 10% of bankroll
test_kelly_bet_zero_when_no_edge	CH5 — p_win ≤ price → bet=0
test_kelly_bet_min_floor	CH5 — applies $1 min when positive edge
test_multi_fire_two_per_window	CH7 — fires twice in same window
test_multi_fire_blocked_at_cap	CH8 — third fire same window returns None
test_multi_fire_resets_on_window_boundary	CH8 — new window resets counter
test_bankroll_getter_integration	CH6 — bet scales with bankroll
test_v2_1_rationale_includes_kelly_metadata	telemetry — Kelly info in rationale

6. Deployment (H.5) + verify

6.1 .envrc change

```
From: export STRATEGY=spot_aware_v2
To: # PHASE-H 2026-05-13: v2.1 deploy
    # export STRATEGY=spot_aware_v2
    export STRATEGY=spot_aware_v2_1
    export KELLY_MULT=0.10
    export MULTI_FIRE_CAP=2
```

Backup pattern: .envrc.phase-H-pre-v2_1-backup-YYYYMMDD-HHMMSS
Rollback (one-line): cp backup .envrc && sudo systemctl restart bot-staging

6.2 Verify cadence

When	Check
+5 min	active strategy: spot_aware_v2_1 in startup log; 0 errors
+5 min	first fire shows new rationale (kelly bet, fires_this_window)
+30 min	service active, no queue overflow, fire rate ≥ projected
+24 h	fires/day ≈ 30 (multi-fire 2× × baseline 16); WR ≥ 47%
+7 d	soak gate per §7 below

7. Soak gate (H.6) — PROMOTE / EXTEND / ROLLBACK

Outcome	Criteria
PROMOTE → Phase I (live trading prep)	WR ≥ 52% on n ≥ 100 AND daily_pnl_min ≥ –$10 AND fires/day ≥ 30 AND queue_overflow
EXTEND (re-soak 7d)	WR 45-52% OR fires/day 20-30 OR daily_pnl in [–30,–10) on worst day
ROLLBACK to v2	WR < 45% over 48h OR daily_pnl < –$30 any day OR bankroll ≤ $500

8. Open questions for operator (Q1-Q5)

These are the implementation knobs the operator decides before H.2 starts.

Q	Question	Options	Recommended
Q1	Hour-of-day gate (S1)?	YES = skip 5/14/23 UTC · NO = no gate	NO (sample too thin)
Q2	Regime gate (W3)?	A no gate · B half-Kelly on chop · C trending-only	A (start simple)
Q3	Multi-fire cap?	2 / 3 per window	2 (conservative)
Q4	Kelly multiplier?	0.10 / 0.25	0.10 (smooth ride)
Q5	Live-trading promotion threshold?	$2k / $5k / $10k	$2k (locked decision #12)

9. What this phase explicitly does NOT do

- Does NOT deploy v2.1 — H.5 is the deploy step (next session, NOT now)
- Does NOT touch v2/v3/v1/wide_fire — v2.1 is additive (new file)
- Does NOT modify any locked decision
- Does NOT re-engage ensemble (Phase F cut intact)
- Does NOT skip the 7-day soak gate before promotion

10. Honest pushback

> **The 58% WR backtest is from a 5-day window.** v2.1's exploitation of the 30-50%×|Δ|<5 cell assumes that cell's edge generalizes. Phase G Scenario C showed: at 50% WR (regression), median still grows but 97% of paths hit catastrophic drawdown.

Mitigations baked into v2.1:

- Kelly 0.10× (not 0.25×) for first 30 days — smoother trajectory
- $500 absolute bet cap — Polymarket liquidity realism
- Multi-fire 2× (not 3×) — limit per-window capital exposure
- 7-day soak gate before promotion — kill switch on bad data

Even with these mitigations: **if WR drops below 45% over 48h, ROLLBACK**. The Kelly variance signature is real.

End of Phase H implementation plan · 2026-05-13 · 14 edges inventoried · 4 filter changes + 4 sizing/multi-fire changes · 6 sub-phases · 1 PDF

Source prompt: phase-prompts/phase-H-v2.1-implementation-2026-05-13.txt · Reference: pdfs/PHASE_G_EDGE_DISCOVERY.pdf